

Computational Economics: Problem Set 4

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Question 1 & 2 & 5

- The code and explanations are in the .py file.

Question 3

Table 1: MLE estimates of the ln-wage model by year (male heads, age 25–60, wage > \$7/hr)

	1971	1980	1990	2000
Constant	0.5863	1.0023	0.2772	-0.2952
Educ	0.0665	0.0660	0.0955	0.1102
Age	0.0649	0.0456	0.0579	0.0845
Age ²	-0.0006	-0.0004	-0.0005	-0.0009
Black	-0.1641	-0.1030	-0.1680	-0.2598
Hispanic	0.0000	0.0000	0.0000	0.0000
OtherRace	0.0175	0.0123	-0.0520	-0.0622
$\hat{\sigma}$ (MLE)	0.4059	0.4478	0.4815	0.5342
Log-likelihood	-713.71	-1142.42	-1385.08	-2043.02
n	1380	1856	2013	2580

- The likelihood function is in the next page.

Estimates from $\ln w_{it} = \alpha + \beta_1 \text{Educ}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Age}_{it}^2 + \beta_4 \text{Black}_{it} + \beta_5 \text{Hispanic}_{it} + \beta_6 \text{OtherRace}_{it} + \varepsilon_{it}$.
White is the baseline (=1). Assumed the homoskedastic and normally distributed error term.

Question 4

Interpret β_1 (Educ), based on year-by-year estimates. How do the returns to education change over time in these data?

- **1971:** $\beta_1 = 0.0665$, $100(e^{0.0665} - 1)$, $\approx 6.9\%$, so each year of additional schooling increases the wage by approximately 6.9%.
- **1980:** $\beta_1 = 0.0660$, $\approx 6.8\%$ per year, so each year of additional schooling increases the wage by approximately 6.8%.
- **1990:** $\beta_1 = 0.0955$, $\approx 10.0\%$ per year, so each year of additional schooling increases the wage by approximately 10.0%.
- **2000:** $\beta_1 = 0.1102$, $\approx 11.7\%$ per year, so each year of additional schooling increases the wage by approximately 11.7%.

Likelihood Function

Let $y \in \mathbb{R}^n$ be $\ln(\text{wages})$, $X \in \mathbb{R}^{n \times k}$ be covariates that shown in the equation of Q3, $\beta \in \mathbb{R}^k$, and $\sigma^2 > 0$. Assume

$$y \mid X \sim \mathcal{N}(X\beta, \sigma^2 I_n).$$

$$L(\beta, \sigma^2; y, X) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} (y - X\beta)^\top (y - X\beta)\right).$$

$$\ell(\beta, \sigma^2; y, X) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (y - X\beta)^\top (y - X\beta).$$

pytest

```
!PYTHONPATH=/content PYTEST_DISABLE_PLUGIN_AUTOLOAD=1 pytest -q tests/test_pset4.py

... [100%]
3 passed in 1.13s
```

Figure 1: Pytest run showing all tests passed

- `tests/test_pset4.py` — pytest test suite
- `econ_833_ps4.py` — main script to run Q1–Q5 (not in pytest format)
- `econ_833_ps4.pytest.py` — writes the test file and runs pytest
- note: my code is written in google colab (other parts will be same but opening the file part might be slightly different that appear in the code)